The Changing Geography of Metropolitan Opportunity: The Segregation of the Poor in U.S. Metropolitan Areas, 1970 to 1990

Presentation

By

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“Poor” and “Rich” Areas

Colloquially, most or us are aware of what is considered to be a “poor area” or a “rich area”. This article strives to quantify how poor and non poor areas are distributed and isolated.





Why study Segregation of “Rich” and “Poor” Areas

- This article analyzes the segregation of the poor in major cities across all parts of the United States and analyzes the trends from 1970-1990.
- In this study, the “poor” are defined as persons that are below the poverty line. Note: the poverty line was not defined in this article.
- This data can be used by policy makers in decisions regarding funding and can be used as indicators on whether current policies are working or not.

Why study Segregation of “Rich” and “Poor” Areas (cont.)

- Separation of poor from the nonpoor has several unfortunate consequences
  - *Isolates them in distressed areas* where the neighborhood effects of living among negative role models may subvert positive individual or family efforts
  - Isolated poor communities may be *inadequately served by underfunded public institutions* that do little help residents break out of distressed neighborhoods.
  - Leave them more susceptible to the harmful effects of economic downturn. They have *fewer neighborhood resources* to call on to relieve dislocations.
  - May reinforce *negative stereotypes of the poor* and weaken the bonds between the poor and nonpoor that allow the two groups to live in relative peace.
  - scarcity of positive role models

Segregation Indices

- The *dissimilarity index* measures the evenness of the distribution of the poor within a metropolitan area. The dissimilarity index describes *the percentage of poor people that would have to move from one census tract to another to achieve an even distribution*. A dissimilarity index of 0 means that the poor are perfectly distributed throughout all census tracts. A dissimilarity index of 100 means that 100 percent of all the poor need to move to achieve an even distribution.

- The isolation index in this article measures the *extent to which the poor are exposed in their neighborhoods to other poor people*. This index varies from 0 to 100. Zero indicates someone completely around people other than poor and 100 indicates complete isolation of the poor from everyone else.

[3] Massey and Denton (1988) give the following formula for dissimilarity:

$$D = \sum_{i=1}^{n} \frac{t_i |p_i - P|}{2TP(1-P)},$$

where t_i and p_i are the total population and minority proportion of areal unit i, and T and P are the population size and minority proportion of the whole city, which is divided into n areal units. We present dissimilarity values as whole numbers rather than fractions in this article (i.e., we multiply D by 100), except in the regression analyses.

[4] Census tracts have an average population of 4,000 and are used by many researchers as an approximation for neighborhoods.

[5] Massey and Denton (1988) give the following formula for isolation:

$$_xP_x^* = \sum_{i=1}^{n} \left[\frac{x_i}{X} \bullet \frac{x_i}{t_i} \right],$$

where x_i and t_i are the numbers of group members and the total population of areal unit i, and X is the number of group members citywide. We present isolation values as whole numbers rather than fractions in this article (i.e., we multiply $_xP_x^*$ by 100), except in the regression analyses.

[6] For further discussion of the indices of dissimilarity and isolation, see Massey and Denton (1988).

Dissimilarity and Isolation of Poor in Selected Cities

In raw numbers, 36.1 percent in 1990 meant that 5.7 million poor people would have to relocate to different neighborhoods within the same metropolitan area. A value of 21.0 on the isolation index indicates that the neighborhood of an average poor person is 21.0 percent poor.

Table 1. **Dissimilarity, Isolation, Number of Poor Persons, and Number of Poor Persons Needing to Move to Eliminate Segregation, 100 Largest U.S. Metropolitan Areas, 1990**

Metropolitan Area	Dissimilarity of Poor	Isolation of Poor	Number of Poor	Number of Poor Needing to Move to Eliminate Segregation
Akron, OH	39.7	22.9	77,383	26,997
Albany-Schenectady-Troy, NY	34.9	15.6	73,524	23,442
Albuquerque, NM	34.0	22.0	78,697	22,820
Allentown-Bethlehem-Easton, PA	34.5	14.3	48,126	15,420
Anaheim–Santa Ana–Garden Grove, CA	35.3	14.3	200,860	64,854
Atlanta, GA	39.6	22.4	279,507	99,572
Austin, TX	37.6	24.8	115,330	36,720
Bakersfield, CA	32.3	23.7	89,157	23,909
Baltimore, MD	46.0	24.3	233,498	96,688
Baton Rouge, LA	36.1	29.2	96,718	28,359
Birmingham, AL	35.5	25.2	135,854	40,873
Wilmington, DE-NJ-MD	34.6	15.5	44,210	14,079
Youngstown-Warren, OH	37.1	24.8	67,120	21,463
Mean	36.1	21.0	170,996	57,023
Total			17,099,588	5,702,278

Source: Based on data in The Urban Institute's Under Class Data Base.

Indices of Poor as Compared with Black and Hispanic Populations

As seen on the table below, the dissimilarity and isolation of the black population is much larger than the average poor person in the U.S. and the Hispanic population (1990).

Race	Dissimilarity	Isolation
Black	60.2	42.2
Hispanic	39.1	17.2
Average Poor in US	36.1	21.0

While segregation of the poor increased, the segregation of blacks decreased. The dissimilarity of blacks fell by 7.2 (11 percent). For non whites, the dissimilarity fell by 15.8 (23 percent) from 1970-1990.

A partial explanation for this is that fair housing laws have permitted blacks and other minorities to move to more integrated suburban areas from poorer areas.

Dissimilarity Indices compared with other known indicators of poorer neighborhoods

One concern that might affect the numbers in this study is what is considered “short term poor”. These are people who are temporarily misplaced because of layoffs or other factors. To alleviate this, the authors compared factors that are considered more permanent indicators of poor such as the number of high school dropouts and Welfare recipients.

Table 2. **Mean Dissimilarity Indices and Correlations with Dissimilarity of the Poor, 100 Largest U.S. Metropolitan Areas, 1990**

Dissimilarity Index	Metropolitan Area Average Dissimilarity	Correlation with Dissimilarity of Poor
Poor	36.1	
High school dropouts	35.2	0.53
Foreign born	29.7	0.11
Female-headed families	30.7	0.81
Welfare recipients	34.3	0.92
Unemployed	22.1	0.79
Limited English proficiency	39.6	0.38
Blue-collar workers	21.4	0.29
Blacks	60.6	0.59
Hispanics	39.1	0.44

Source: Based on data in The Urban Institute's Under Class Data Base.

Metro areas with highest Segregation Indices

The segregation indices varied greatly among metropolitan areas. Dissimilarity index range was from 19.6 (Bristol TN-VA) to 55.1 (Milwaukee). The Isolation Index ranged from 7.1 (Nassau-Suffolk, NY) to 35.4 (El Paso, TX).

Table 3. **Hundred Largest U.S. Metropolitan Areas Ranked by Dissimilarity of the Poor, Top and Bottom 10 Areas, 1990**

Rank	Metropolitan Area	Dissimilarity of Poor
1	Milwaukee, WI	55.1
2	Hartford, CT	52.8
3	Cleveland, OH	51.4
4	Detroit, MI	50.1
5	Chicago, IL	49.8
6	Newark, NJ	48.2
7	Philadelphia, PA-NJ	47.9
8	Baltimore, MD	46.0
9	New Haven–West Haven, CT	46.0
10	Buffalo, NY	44.8
91	Las Vegas, NV	28.5
92	Tacoma, WA	27.8
93	Knoxville, TN	27.3
94	Orlando, FL	27.2
95	Portland, OR-WA	27.1
96	Vallejo-Fairfield-Napa, CA	25.5
97	Riverside–San Bernardino–Ontario, CA	24.7
98	Jersey City, NJ	24.0
99	Northeast Pennsylvania, PA	20.6
100	Johnson City–Kingsport–Bristol, TN-VA	19.6

Source: Based on data in The Urban Institute's Under Class Data Base.

Table 4. **Hundred Largest U.S. Metropolitan Areas Ranked by Isolation of the Poor, Top and Bottom 10 Areas, 1990**

Rank	Metropolitan Area	Isolation of Poor
1	El Paso, TX	35.4
2	New Orleans, LA	35.2
3	Memphis, TN-AR-MS	33.4
4	Mobile, AL	32.3
5	Milwaukee, WI	32.3
6	San Antonio, TX	30.0
7	Cleveland, OH	29.9
8	Fresno, CA	29.6
9	New York, NY-NJ	29.4
10	Baton Rouge, LA	29.2
91	Honolulu, HI	14.0
92	Paterson-Clifton-Passaic, NJ	13.6
93	Northeast Pennsylvania, PA	13.6
94	Seattle-Everett, WA	13.1
95	San Jose, CA	12.3
96	Long Branch–Asbury Park, NJ	11.4
97	Oxnard–Simi Valley–Ventura, CA	11.2
98	Vallejo-Fairfield-Napa, CA	10.1
99	New Brunswick–Perth Amboy–Sayreville, NJ	10.0
100	Nassau-Suffolk, NY	7.1

Source: Based on data in The Urban Institute's Under Class Data Base.

Change in Dissimilarity and Isolation Indices of the Poor for Selected Metro Areas 1970-1990

Table 7. **Dissimilarity of the Poor and Change in Dissimilarity of the Poor, 100 Largest U.S. Metropolitan Areas, 1970–1990**

Metropolitan Area	Dissimilarity of Poor 1970	1980	1990	Change, 1970–90
Akron, OH	32.6	38.3	39.7	7.1
Albany-Schenectady-Troy, NY	27.4	32.8	34.9	7.6
Albuquerque, NM	36.2	32.8	34.0	–2.2
Allentown-Bethlehem-Easton, PA	23.6	29.5	34.5	10.9
Anaheim–Santa Ana–Garden Grove, CA	22.7	28.6	35.3	12.6
Atlanta, GA	39.8	38.5	39.6	–0.2
Austin, TX	34.1	37.1	37.6	3.5
Bakersfield, CA	34.2	29.1	32.3	–1.9
Baltimore, MD	40.1	45.3	46.0	6.0
Baton Rouge, LA	39.5	36.9	36.1	–3.4
Birmingham, AL	29.1	34.8	35.5	6.4
Boston, MA	31.2	35.7	37.4	6.2
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Vallejo-Fairfield-Napa, CA	19.5	25.4	25.5	6.0
Washington, DC-MD-VA	37.2	39.8	38.1	0.9
West Palm Beach–Boca Raton, FL	35.0	35.0	34.4	–0.6
Wichita, KS	32.2	31.7	33.9	1.7
Wilmington, DE-NJ-MD	34.7	32.4	34.6	–0.1
Youngstown-Warren, OH	32.8	35.6	37.1	4.3
Mean*	32.9	34.8	36.4	3.5

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: N/A refers to data unavailable for 1970.
* All means exclude five metropolitan areas missing data for 1970.

Table 8. **Isolation of the Poor and Change in Isolation of the Poor, 100 Largest U.S. Metropolitan Areas, 1970–1990**

Metropolitan Area	Isolation of Poor 1970	1980	1990	Change, 1970–90
Akron, OH	15.2	17.8	22.9	7.8
Albany-Schenectady-Troy, NY	13.0	16.0	15.6	2.6
Albuquerque, NM	25.9	20.5	22.0	–3.9
Allentown-Bethlehem-Easton, PA	10.8	12.6	14.3	3.6
Anaheim–Santa Ana–Garden Grove, CA	8.6	10.9	14.3	5.8
Atlanta, GA	23.5	24.3	22.4	–1.1
Austin, TX	25.9	23.8	24.8	–1.1
Bakersfield, CA	24.1	17.7	23.7	–0.4
Baltimore, MD	23.0	26.1	24.3	1.3
Baton Rouge, LA	29.8	23.8	29.2	–0.6
Birmingham, AL	25.9	24.7	25.2	–0.7
Boston, MA	14.9	16.9	16.2	1.3
Buffalo, NY	15.5	20.5	24.9	9.5
Charleston–North Charleston, SC	35.7	24.8	23.5	–12.2
Charlotte-Gastonia, NC	23.1	18.8	17.0	–6.1
Chattanooga, TN-GA	27.1	22.6	21.8	–5.3
Chicago, IL	20.2	27.6	28.7	8.5
Wichita, KS	17.1	15.3	18.4	1.3
Wilmington, DE-NJ-MD	18.0	19.6	15.5	–2.5
Youngstown-Warren, OH	14.5	18.3	24.8	10.3
Mean*	19.5	19.7	21.3	1.8

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: N/A refers to data unavailable for 1970.
* All means exclude five metropolitan areas missing data for 1970.

Change in Dissimilarity and Isolation Indices of the Poor for 1970-1990 – Top Metro Areas

Milwaukee experienced the largest increase in the dissimilarity of the poor (16.3) and isolation of the poor (15.3). The largest decrease was Tulsa (-6.0). The largest decrease of isolation was Charleston, SC (-12.2)

Table 10. **Hundred Largest U.S. Metropolitan Areas Ranked by Change in Dissimilarity of the Poor, Top and Bottom 10 Areas, 1970–1990**

Rank	Metropolitan Area	Change in Dissimilarity of Poor
1	Milwaukee, WI	16.3
2	Hartford, CT	13.9
3	Syracuse, NY	12.8
4	Anaheim–Santa Ana–Garden Grove, CA	12.6
5	Buffalo, NY	11.7
6	Allentown-Bethlehem-Easton, PA	10.9
7	Springfield-Chicopee-Holyoke, MA-CT	10.9
8	Detroit, MI	10.8
9	Gary–Hammond–East Chicago, IN	9.8
10	New Haven–West Haven, CT	9.8
86	Knoxville, TN	–2.9
87	Tacoma, WA	–2.9
88	Baton Rouge, LA	–3.4
89	Nashville-Davidson, TN	–3.7
90	Jacksonville, FL	–4.2
91	Charleston–North Charleston, SC	–4.9
92	Chattanooga, TN-GA	–5.1
93	Orlando, FL	–5.8
94	Charlotte-Gastonia, NC	–5.9
95	Tulsa, OK	–6.0

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Table excludes five metropolitan areas missing data for 1970.

Table 11. **Hundred Largest U.S. Metropolitan Areas Ranked by Change in Isolation of the Poor, Top and Bottom 10 Areas, 1970–1990**

Rank	Metropolitan Area	Change in Isolation of Poor
1	Milwaukee, WI	15.3
2	Flint, MI	13.3
3	Detroit, MI	12.4
4	Springfield-Chicopee-Holyoke, MA-CT	10.5
5	Youngstown-Warren, OH	10.3
6	Buffalo, NY	9.5
7	Gary–Hammond–East Chicago, IN	9.4
8	Cleveland, OH	9.2
9	Hartford, CT	8.8
10	Lansing–East Lansing, MI	8.6
86	Norfolk–Virginia Beach–Portsmouth, VA	–4.4
87	Greensboro–Winston-Salem–High Point, NC	–4.5
88	Knoxville, TN	–4.5
89	Chattanooga, TN-GA	–5.3
90	West Palm Beach–Boca Raton, FL	–5.4
91	Charlotte-Gastonia, NC	–6.1
92	Orlando, FL	–6.4
93	Columbia, SC	–8.0
94	Jacksonville, FL	–8.1
95	Charleston–North Charleston, SC	–12.2

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Table excludes five metropolitan areas missing data for 1970.

Regression Analysis of Selected Metro Area Characteristics and Dissimilarity of the Poor

- The dissimilarity of the poor in 1990 is higher in areas that experienced larger increases in poverty rate.
- The poor are more segregated in areas with larger percentages of blacks in the population. Thus the level of black segregation in an area is apt to have some effect on the level of segregation of the poor in the area.
- The isolation of the poor in 1990 is significantly related to 1990 poverty rate.
- The are poor are significantly more isolated in the Midwest are than the North-east areas.

Table 5. **Regression of Selected Metropolitan Area Characteristics on Dissimilarity of the Poor, 100 Largest U.S. Metropolitan Areas, 1990**

Independent Variable	Coefficient	*t* Statistic
Intercept	0.358	18.49***
1990 population	5×10^{-9}	1.50
Percent change in population, 1970–90	–0.0002	–0.23
1990 poverty rate	–0.226	–1.34
Change in poverty rate, 1970–90	0.533	2.05*
Black share of population, 1990	0.372	4.79***
South region dummy	–0.051	–2.86**
Midwest region dummy	0.022	1.37
West region dummy	–0.044	–2.86**
Regression statistics		
Adjusted R^2	0.49	
F value for regression	12.78	
Observations	100	
Mean of dependent variable	0.361	

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Default region is Northeast. Dependent variable is dissimilarity of the poor, 1990.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 6. **Regression of Selected Metropolitan Area Characteristics on Isolation of the Poor, 100 Largest U.S. Metropolitan Areas, 1990**

Independent Variable	Coefficient	*t* Statistic
Intercept	0.062	6.09***
1990 population	2×10^{-9}	1.03
Percent change in population, 1970–90	–0.0002	–0.35
1990 poverty rate	1.110	12.47***
Change in poverty rate, 1970–90	0.222	1.62
Black share of population, 1990	0.198	4.84***
South region dummy	–0.020	–2.15*
Midwest region dummy	0.019	2.24*
West region dummy	–0.025	–3.03**
Regression statistics		
Adjusted R^2	0.81	
F value for regression	53.01	
Observations	100	
Mean of dependent variable	0.210	

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Default region is Northeast. Dependent variable is isolation of the poor, 1990.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Regression Analysis of Selected Metro Area Characteristics on Change in Dissimilarity or the Poor

- Larger increase in the dissimilarity of the poor from 1970-1990 occurred in metropolitan areas with lower 1970 poverty rates (table 12). More mixing of poor and nonpoor occur when the poor start off as a larger share of the population.
- Larger increases in the isolation of the poor from 1970-1990 occurred in metropolitan areas that experienced larger 1970-1990 increases in poverty rate (table 13)

Table 12. **Regression of Selected Metropolitan Area Characteristics on Change in Dissimilarity of the Poor, 100 Largest U.S. Metropolitan Areas, 1970–1990**

Independent Variable	Coefficient	*t* Statistic
Intercept	0.098	6.43***
1970 population	-1×10^{-9}	–0.44
Percent change in population, 1970–90	–0.020	–1.77
1970 poverty rate	–0.325	–2.52*
Change in poverty rate, 1970–90	0.082	0.49
Black share of population, 1970	0.064	0.94
South region dummy	–0.047	–3.16**
Midwest region dummy	–0.006	–0.50
West region dummy	–0.019	–1.55
Regression statistics		
Adjusted R^2	0.47	
F value for regression	10.70	
Observations	90	
Mean of dependent variable	0.034	

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Default region is Northeast. Regression excludes 10 metropolitan areas with data missing for 1970. Dependent variable is change in dissimilarity of the poor, 1970–90.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 13. **Regression of Selected Metropolitan Area Characteristics on Change in Isolation of the Poor, 100 Largest U.S. Metropolitan Areas, 1970–1990**

Independent Variable	Coefficient	*t* Statistic
Intercept	0.032	3.50***
1970 population	4×10^{-10}	0.21
Percent change in population, 1970–90	–0.006	–0.84
1970 poverty rate	–0.093	–1.21
Change in poverty rate, 1970–90	1.184	11.89***
Black share of population, 1970	0.064	1.57
South region dummy	–0.022	–2.45*
Midwest region dummy	0.011	1.57
West region dummy	–0.015	–2.04*
Regression statistics		
Adjusted R^2	0.82	
F value for regression	51.91	
Observations	90	
Mean of dependent variable	0.017	

Source: Based on data in The Urban Institute's Under Class Data Base.
Note: Default region is Northeast. Regression excludes 10 metropolitan areas with data missing for 1970. Dependent variable is change in isolation of the poor, 1970–90.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Recommended Use of Results

In an effort to decrease the segregation of the poor, the authors recommend a broad range of policy options.

- A set of programs that would attempt to attract the nonpoor into poor neighborhoods through community redevelopment programs.
- Provide transportation assistance to jobs in the suburbs or by finding them housing in nonpoor areas.
- Focus on the overall health of metropolitan-area or regional economies that might eventually affect the level of segregation

Critique

- Statistical analysis seems to be very sound for most of the results of this article.
- The authors did make their case by approaching the findings in a variety of ways and comparing their results with other studies.
- The authors' results did match their data and was very understandable.
- It would have been nice to see other correlations such as crime rate affects the cities with the highest and lowest segregation of the poor.
- It would an interesting next step to examine the factors going into why the indices of these cities changed between 1970 and 1990.

Questions?